

Retail Performance Analytics Showcase

Cloud Data Pipeline Implementation: Databricks, SQL & Power BI



Project Overview

Architecting an end-to-end analytics solution to transform global sales data into business intelligence.

Business Context

The Challenge

Retail management required real-time visibility into month-over-month growth trends across global segments. Data was siloed in legacy flat files with inconsistent date formatting and schema errors.

The Solution

Built a rapid-deployment pipeline in **Databricks** to clean, model, and aggregate sales data. Developed an interactive **Power BI** dashboard for executive performance auditing.

Modern Data Tech Stack



Databricks

Managed Spark environment utilized for large-scale data ingestion and automated schema inference.



SparkSQL

Implementation of **LAG Window Functions** and CTEs to calculate temporal growth metrics.



Power BI

Deployment of a high-fidelity dashboard with category drill-downs and interactive slicers.



Clean Code

Version-controlled SQL scripts and reproducible notebooks stored on GitHub for collaborative visibility.

Medallion Data Pipeline

- **Bronze (Raw):** Direct ingestion of Superstore datasets into Delta tables.
- **Silver (Clean):** Data type casting using `to_date` and `try_cast` for malformed strings.
- **Gold (Aggregated):** Feature-engineered views optimized for high-performance BI reporting.



| Market Analysis Impact

2.3M

Total Revenue Analyzed

Aggregated sales across four global regions, identifying critical seasonal volatility in high-volume sectors.

Revenue by Segment



Technology leads in volume but exhibits the highest volatility in growth patterns.

Growth Performance Trend



Advanced period-over-period auditing reveals a 120% surge in Q4 Technology sales.

Segment Efficiency Audit

Category	Avg. Growth	Total Profit	Market Status
Technology	24.5%	\$145,454	High Growth
Office Supplies	12.2%	\$122,490	Stable
Furniture	8.1%	\$18,451	Low Margin

Hamburg Context

Applying these analytics frameworks to the **Hamburg Metropolitan Region** enables logistics firms to optimize supply chains by correlating regional inventory with real-time sales velocity.

This project demonstrates readiness to handle local data challenges at scale.



Portfolio Complete

Thank you for your attention. I am looking forward to discussing how my skills in cloud analytics can add value to your team.

```
SELECT * FROM hamburg_candidates WHERE skill = 'SQL/Databricks';
```

| Image Sources

1. <https://dtyped.com/demystifying-the-medallion-architecture/>
2. <https://media.istockphoto.com/id/507372675/de/foto/hamburger-rathaus-und-alster-lake.jpg?s=612x612&w=0&k=20&c=ILldPoUTYVkMYw54UB1Auas1XoeOQCdqCdybz-qtWI=>